

Index

Abstract	v
Index	v
List of Figures	vi
List of Tables	vii
Nomenclature	viii
1 Introduction	1
1.1 Introduction	1
1.2 Motivation	2
1.3 Objective	3
1.4 Problem Statement	3
1.5 Approach	4
1.6 Scope of the Project	4
1.7 Organization of the Rest of the Report	5
2 Literature Review	6
2.1 Previous Approaches to Solve the Problem	6
2.1.1 Existing Collection-Management Products	6
2.1.2 Conversational Voice Agents	6
2.1.3 The WhatsApp Multi-Device Protocol	7
2.1.4 Anti-Spam and Rate-Limiting Heuristics	7
2.1.5 Account Aggregator Framework	8
2.2 Problem Identification	8
3 Proposed Methodology and Implementation	10
3.1 Development Methodology	10
3.2 High-Level Architecture	10
3.3 Case Management	11
3.4 The Workflow Engine	12
3.5 The Campaign Module	13
3.6 AI Voice Calling	14
3.7 WhatsApp Automation	15
3.8 SMS via Android	17
3.9 The Desktop Application	17
3.10 Data Analysis on Customer Records	17
3.11 Summary of Contributions	18
4 Software Design	19
4.1 Design Goals	19
4.2 Component Decomposition	20
4.3 Data Model	21
4.4 Principal Flow: AI Voice Call	21
4.5 Security Design	22
5 Results and Discussion	23
5.1 Outcomes of the Proposed Platform	23
5.2 Channel-Level Contact Rates	23
5.3 Speed Dialer Throughput	25
5.4 WhatsApp Sender Pool Behaviour	26
5.5 AI Voice Agent Performance	27
5.6 System-Level Metrics	28
5.7 Failure Modes and Stakeholder Feedback	28
6 Conclusions and Future Scope	30
6.1 Conclusion	30
6.2 Reflections on What Could Have Been Done Better	30
6.3 Future Scope	31
6.4 Closing Note	31
References	31
Appendix A: Selected Platform Capabilities	35
Appendix B: Campaign Upload Schema	36